

TITLE PAGE

- **Problem Statement ID – SIH26007**
- **Problem Statement Title – Safe and Efficient Operation of Mine Vehicles in Fog and Low Visibility Conditions in Open Cast Iron Ore Mines**
- **Theme – Smart Automation/Hardware**
- **PS Category – Hardware**
- **Team ID – SIH26-A0H-T350**
- **Team Name (Registered on portal) – Pixels**



PROPOSED SOLUTION



PROBLEM AT HAND



Dense Monsoon Fog

Bailadila haul roads face severe fog/low visibility, forcing dumpers to slow down or stop.



Single-Sensor Fragility

Vision-only perception fails outright once RGB visibility degrades.



Unpredictable Hazards

Other dumpers, workers, animals & obstacles appear on unmarked haul roads.



Edge Compute Constraints

Real-time, explainable risk decisions must run on low-cost, low-power hardware.



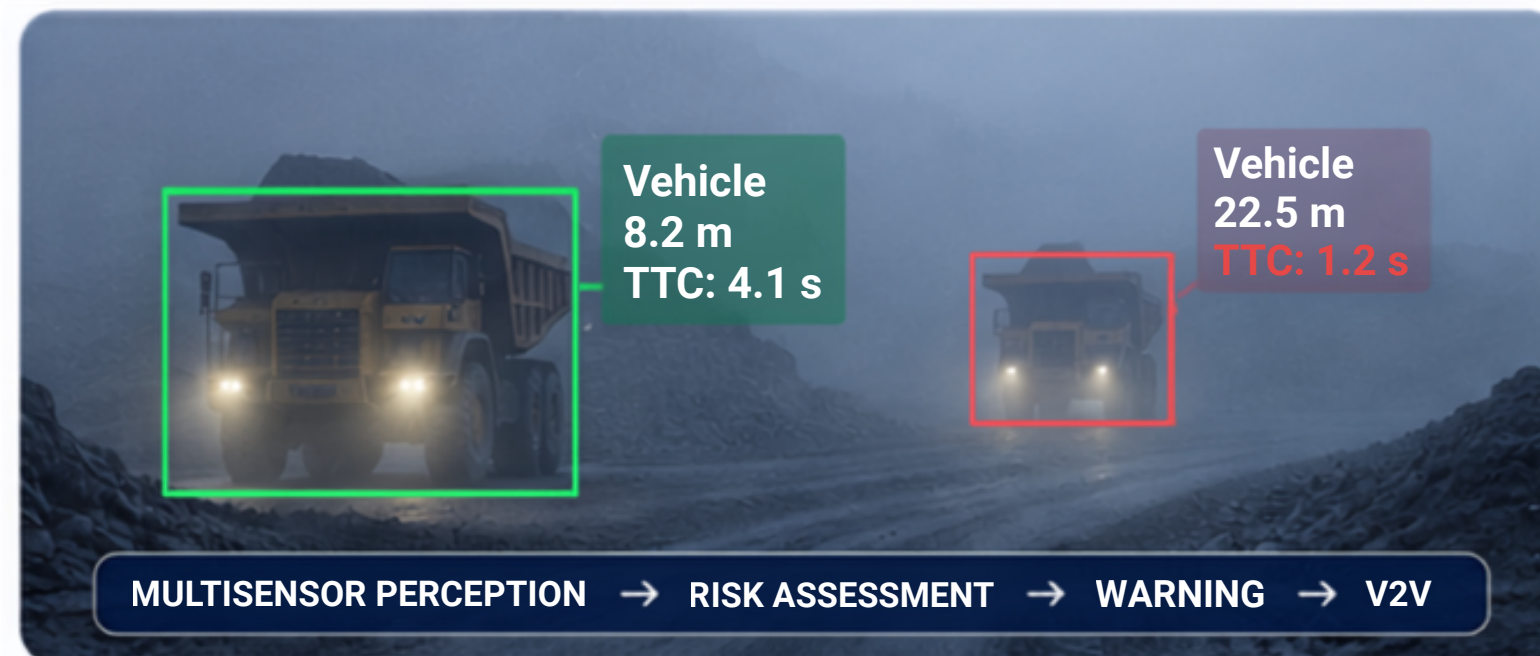
Mine's Zero Visibility Case

When the road disappears, safety can't.



OUR SOLUTION

A low-cost multisensor edge safety layer: RGB (YOLOv8n) + MLX90640 thermal confirmation + LD2450 radar, fused into an explainable **Time-to-Collision (TTC)** risk engine — warning operators and other vehicles via V2V, even when a sensor fails.



WHY WE STAND OUT

- ✓ **Mine-specific, not generic ADAS** — purpose-built for open-cast iron-ore haul roads & HEMM operations.
- ✓ **Low-cost homogeneous sensing** — RGB + thermal array + short-range radar, instead of one expensive sensor.
- ✓ **Explainable safety logic** — **TTC (Time To Collision)**, range & closing speed drive every warning.
- ✓ **Graceful degradation** — local warning never depends on CAN/J1939 or V2V being available.
- ✓ **Clear production upgrade path** — every low-cost prototype component maps to a validated production-grade part.



KEY FEATURES



TTC (Time To Collision)

Range + relative velocity → explainable Green / Yellow / Red risk state.



MLX90640 Thermal Confirmation

Independent warm-target detection when RGB visibility collapses.



HLK-LD2450 Radar Tracking

Deterministic parser + tracker for target position & motion.



Ultralytics

YOLOv8n Edge Detection

3.2M-param lightweight semantic detector tuned for Raspberry Pi 5.



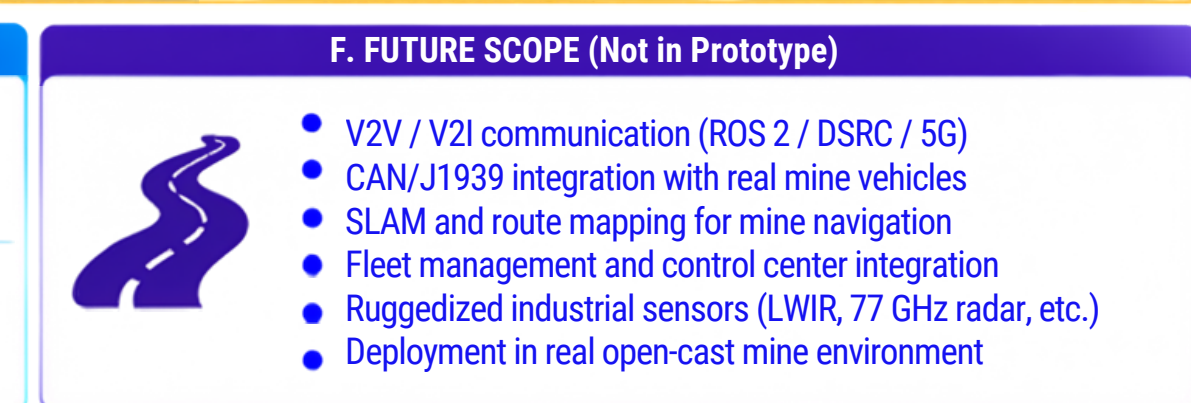
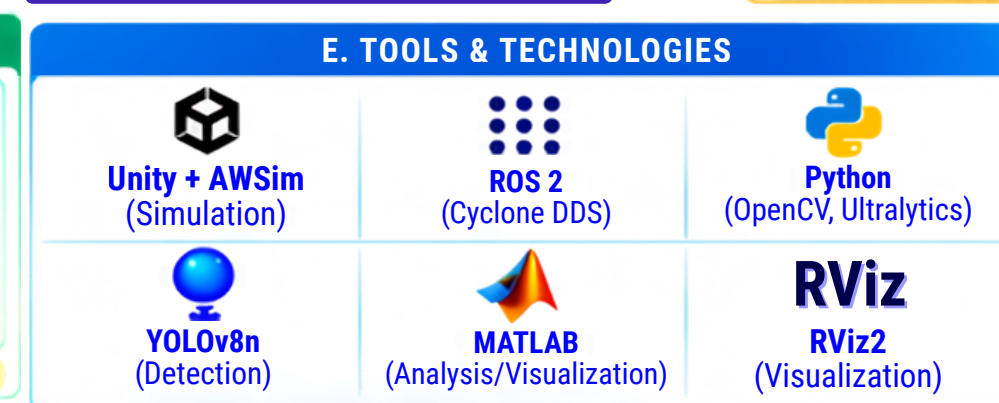
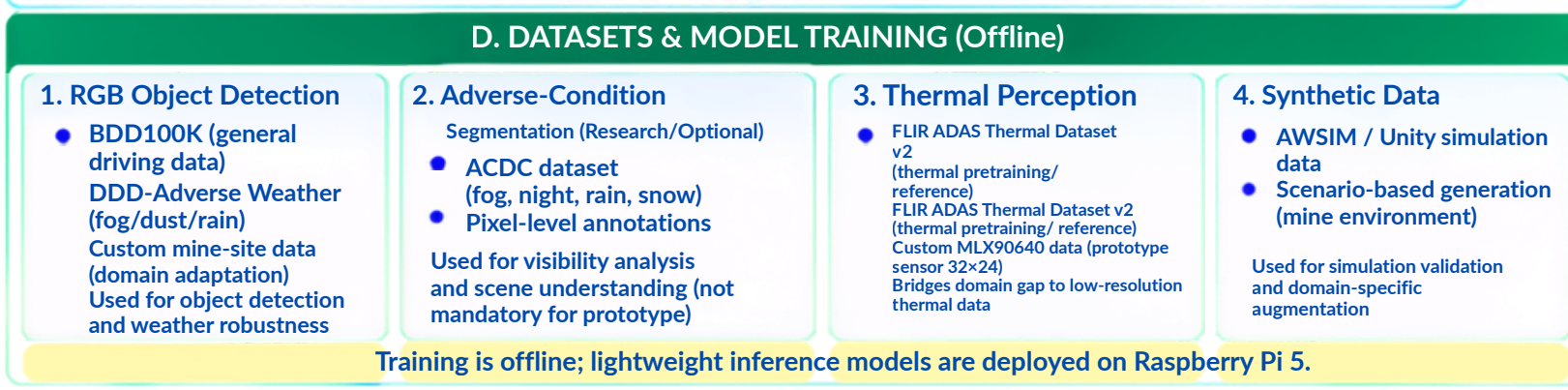
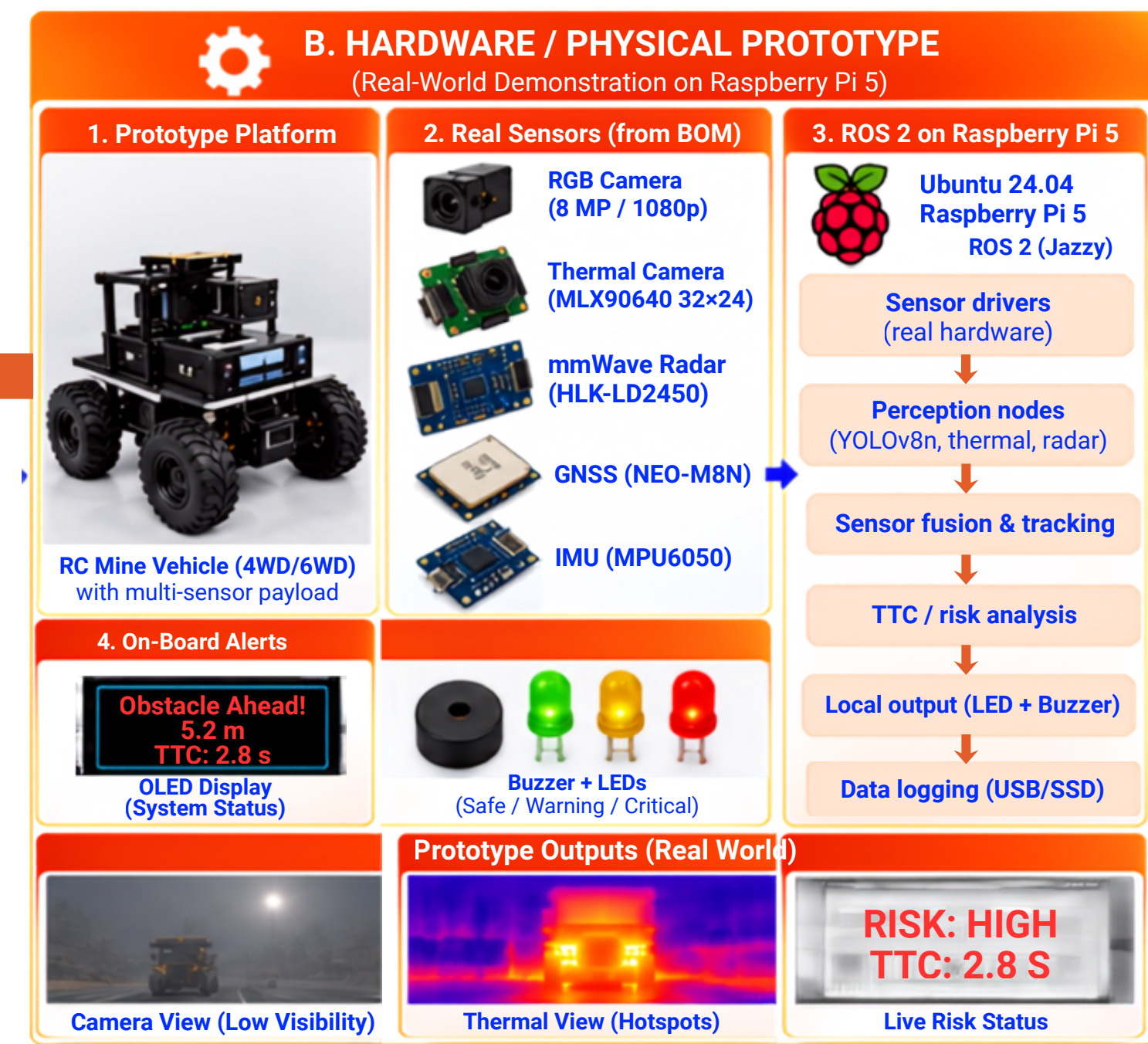
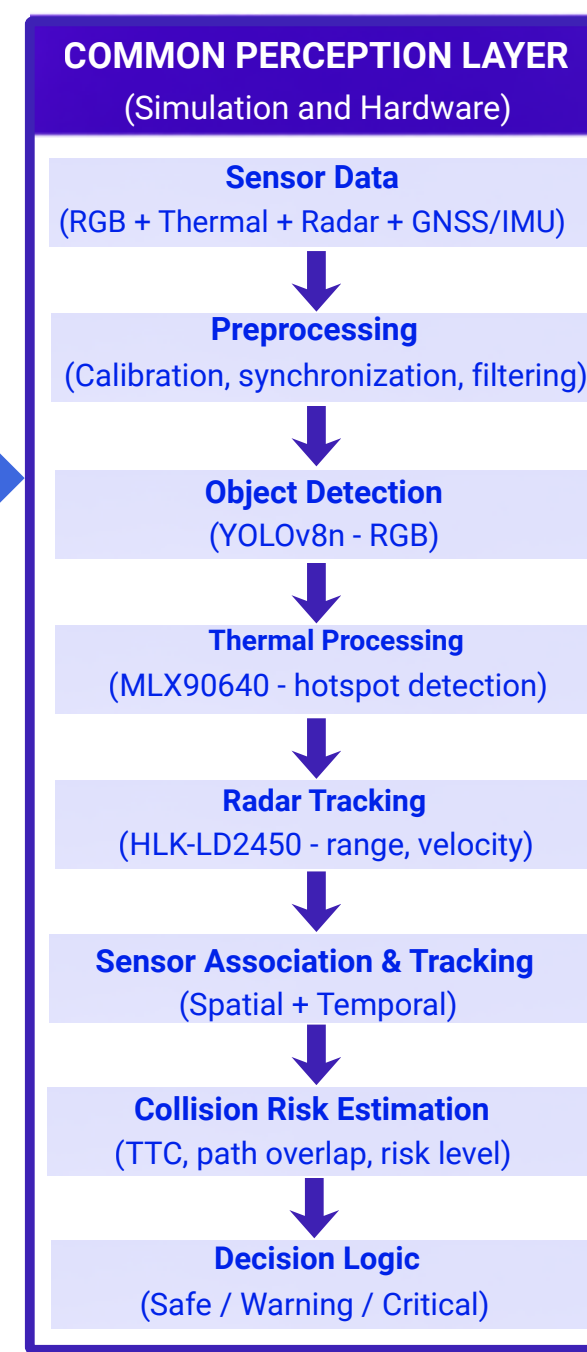
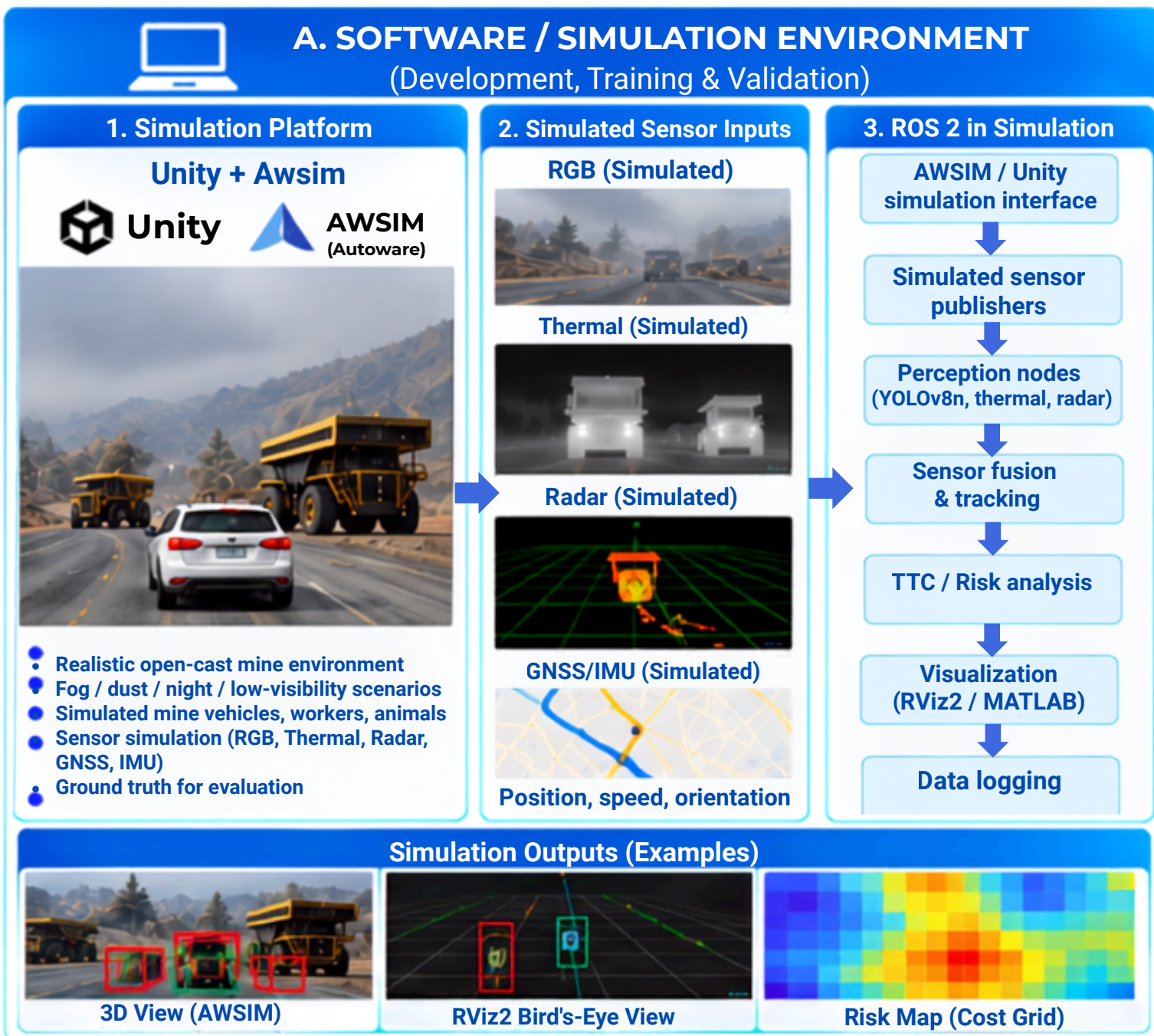
Graceful Degradation

System stays live and warns even if any single sensor drops out.



V2V (ROS 2) with CAN Extendability

- V2V implemented using ROS 2 nodes (e.g., hazard publisher/subscribers).
- Exchange nearby vehicle risk information (position, speed, TTC, hazard state).
- Designed for CAN/J1939 integration in real HEMM (via isolated gateway) for vehicle-state sharing.



TECHNICAL FEASIBILITY

- **Stack:**
Unity-AWSIM (Simulation) $\leftarrow$ ROS 2 $\rightarrow$ Raspberry Pi 5 (Hardware)
- **AI:**
YOLOv8n — 3.2M parameters | 8.7 GFLOPs @ 640×640
- **Sensor Roles:**
RGB / YOLOv8n $\rightarrow$ What
MLX90640 $\rightarrow$ Thermal Confirm
LD2450 $\rightarrow$ Where + Motion
- **Real-Time:**
Thermal O(768) | Radar O(K), $K \leq 3$ | Fusion/TTC O(K) or O(D×K)
- **Target:**
Sub-second warning response on,
final latency benchmarked on Pi 5.

MAINTENANCE & SAFETY FEASIBILITY

- **Graceful Degradation:**
risk loop continues with reduced confidence if a sensor drops.
- **Fail-Safe Warning:**
local SAFE / WARNING / CRITICAL logic is independent of V2V, cloud or CAN.
- **Core BOM:**
Pi 5 4GB - RGB camera - MLX90640 - LD2450 - NEO-M8N
MPU6050 - RC chassis - Motors - motor driver - battery 5V $\geq 5A$ buck - buzzer + LEDs - OLED - USB hub - wiring - mounts - enclosure.
- **Cost:**
Procurement ₹20,876 | Replacement value ₹37,857.

ECONOMIC FEASIBILITY

Parameter	Production Path	MineVision Prototype
Sensing	Industrial LWIR / 77GHz radar / qualified cameras	RGB + MLX90640 + LD2450
Compute	Industrial automotive-grade ECU	Raspberry Pi 5, 4GB
Validation	HEMM field trials + industrial deployment validation	RC-scale + AWSIM + controlled fog tests
Integration	Vehicle CAN/J1939 + production safety ECU	GNSS/IMU now $\rightarrow$ CAN/J1939 gateway later
Deployment risk	Industrial integration & validation complexity	Low-cost proof-of-concept first

Prototype procurement: ₹20,876 | Replacement-value BOM: ₹37,857
Some prototype components are being utilised from our college laboratory resources.

IMPLEMENTATION VIABILITY

Factor	Support / Approach
Manufacturability	Off-the-shelf modules; easy assembly on a small RC chassis
Component availability	All parts sourced via Indian/global electronics suppliers
Serviceability	Modular sensor nodes = quick swap without redesign
Deployment	11-phase roadmap: bring-up $\rightarrow$ validated demo

ENVIRONMENTAL & OPERATIONAL VIABILITY

- **NMDC Context:** Bailadila / Bacheli / Kirandul open-cast iron-ore operations.
- **Environment:** Monsoon, dense fog, high humidity and degraded visibility.
- **Vehicle Fit:** Haul-road awareness around large **Heavy Earth Moving Machinery (HEMM)**.
- **Safety Need:** Blind zones, crossings, workers, animals and obstacles.
- **Adaptation:** Thresholds and sensor mounting calibrated to mine geometry, speeds and site procedures.

SCALABILITY

- **Vehicle-Independent Core:** risk engine needs only a vehicle-state interface swap for real HEMM deployment.
- **Fleet-Ready V2V:** same hazard-message design extends from prototype nodes to a mine fleet.
- **Production Upgrade:**
MLX90640 $\rightarrow$ industrial LWIR;
LD2450 $\rightarrow$ automotive 77GHz radar;
Pi 5 $\rightarrow$ industrial edge ECU.
- **Path:**
Prototype $\rightarrow$ controlled fog testing $\rightarrow$ HEMM pilot $\rightarrow$ V2V.

SOCIETAL IMPACT

Safer Haul Roads

Reduces collision risk for operators, workers, and other vehicles in fog.

Human & Animal Coexistence

Thermal + radar confirmation flags animals/people even in degraded visibility.

Operator Confidence

Explainable Green/Yellow/Red cues reduce fog-driven anxiety and hesitation.

Inclusive Safety

Low-cost design makes advanced safety feasible for more mine sites, not just flagship ones.

OPERATIONAL & ECONOMIC IMPACT

Shorter Haul Cycles

Fewer forced stops/slowdowns in fog sustains ore evacuation throughput.

Lower Accident/Downtime Cost

Early warning reduces collision-related repair and stoppage costs.

Low-Cost Retrofit Path

~₹21K prototype BOM shows a credible low-capital route to fleet-wide safety.

Scalable Fleet V2V

One hazard event can warn every connected vehicle nearby, not just one operator.

ENVIRONMENTAL & STRATEGIC IMPACT

Reduced Idling

Smoother, more confident fog-time operation cuts unnecessary stop-start idling.

Simulation-First Validation

AWSIM testing reduces resource-heavy repeated physical trials.

Future Digital-Twin Ready

Logged sensor + risk data forms the base for mine-scale monitoring later.

National Steel Priority

Directly supports Ministry of Steel / NMDC's open-cast safety mandate.

BENEFITS TO STAKEHOLDERS

NMDC / Mine Operators

Fewer incidents, protected fleet, sustained ore evacuation

Dumper Operators

Reliable fog-time situational awareness and early warning

Ground Workers

Reduced risk of being missed by an approaching vehicle in fog

Ministry of Steel

Supports national open-cast mine safety & productivity goals

Local Communities

Lower accident risk on mine-adjacent haul routes



TOOLS & METHODS

KEY ALGORITHMS, MODELS AND SOFTWARE



YOLOv8n

- YOLOv8 variants (n/s/m/l/x) on parameter count and FLOPs
- Selected YOLOv8n as primary lightweight model for Raspberry Pi 5



Sensor Fusion

- Nearest-neighbour association
- EMA / lightweight Kalman filter
- Multi-sensor tracking



Risk Assessment (TTC)

- $TTC = \text{range} / \text{closing speed}$ (when closing speed > 0)
- Green / Yellow / Red risk states

ROS 2

- ROS 2 Humble middleware
- Sensor, perception, fusion, risk and warning nodes



Unity-AWSIM

- Synthetic low-visibility/fog scenarios for testing and validation

Unity



RESEARCH SUMMARY

KEY FINDINGS AND JUSTIFICATION

- Evaluated YOLOv8n vs YOLOv8s/m/l/x based on parameter count and FLOPs; selected YOLOv8n as the primary lightweight edge model for Raspberry Pi 5 deployment.
- Investigated MLX90640's 32×24 (768-sample) resolution and selected it as a thermal confirmation channel rather than a primary semantic detector.
- Assessed HLK-LD2450 as a deterministic short-range tracker, since research radar-AI datasets such as RADIATE, CARRADA and RADIAL do not match its output format.
- Reviewed FLIR ADAS, LLVIP and radar datasets as reference sources; exact-sensor team-collected data is required for final prototype validation.
- Reviewed SAE J1939 as the reference standard for future heavy-duty HEMM vehicle-state integration.



TECHNICAL REFERENCES

KEY DOCUMENTS, DATASETS AND STANDARDS



YOLOv8 Documentation

Model architecture and performance comparison



FLIR ADAS

Thermal perception research reference

Under confirmation

LLVIP

LLVIP

Visible/infrared low-light research reference

Under confirmation



RADATE / CARRADA / RADIAL

Radar research references



SAE J1939/71

Vehicle application layer (future HEMM integration)



PAPERS

KEY RESEARCH PAPERS USED

- 1. RADIATE — Nobis et al., 2019/2021
- Radar perception in adverse weather; relevant to fog/rain object detection.
- 2. LLVIP — Jia et al., 2021
- Visible-infrared paired dataset for low-light perception and fusion.
- 3. Object Detection Using Near and Far Infrared Imaging — El Maadi et al., 2023
- A systematic literature review on object detection using near and far-infrared imaging: CNN models, challenges, and datasets.